Does a tyre-degradation model transfer?

What public timing cannot identify, and what teams' pit stops reveal

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Abstract

Race-strategy models are fitted and validated inside one championship. I ask two validation questions of one protocol across 7 car classes in Formula 1, WEC, IMSA and ELMS: does a fitted tyre-degradation slope predict a season it never saw, and does a time-optimal planner stop where teams stopped? Transfer is scored by the within-stint R^2, the share of the lap-to-lap change inside a held-out stint that the fitted slope explains. First, public timing rarely carries enough signal to answer the transfer question: inside a stint fuel burn nearly cancels tyre wear, so even a slope stable by construction could not reach 0.2 at 16 of 22 Formula 1 circuits; estimated directly, the slope varies between races by a median 33% of its mean. Second, replaying 1,136 real first stops with each team's next stop held fixed, the planner lands a median of +3 laps [+2, +4] from the Formula 1 team out of sample, but a rule that only balances tyre age before the team's next stop does as well. Third, the stops carry what the fit misses. Read as an implied wear ratio, the ratio of the first stint to the next is 0.73 [0.68, 0.77] when teams move to a harder compound and 1.74 [1.43, 2.53] when they move to a softer one, in the direction hardness predicts for 79% of decisions, against 48% for the fitted slopes: teams act on a compound difference the fitted slopes resolve only in part.

Keywords: race strategy; tyre degradation; out-of-sample validation; motorsport analytics; Formula 1; endurance racing; pit stop optimisation; retrospective audit

1 Introduction

A pit stop is a decision under uncertainty with a measurable cost, and a literature of optimisers has grown around it. This paper does not add another. It asks two validation questions that the existing work, for understandable reasons, mostly does not ask, and answers them with one protocol applied to four championships.

The first question is *transfer*: does a degradation slope fitted on past seasons of a circuit predict the next season there? The second is *agreement with practice*: does a planner built on those slopes stop where a professional pit wall stopped? The second is not a question about which plan was faster. It measures where a model and a practice part company, and by how much.

Three contributions come out of it.

Transfer is mostly not measurable from public timing, and the reason is physical. Inside a stint, tyre age and the fuel burned advance together, so the lap-to-lap shape a stint shows is the net of wear and fuel gain, and the two nearly cancel. A noise ceiling, the score a slope that is stable by construction earns against the lap noise actually present, falls below the threshold of 0.2 used here at most circuits. Estimated directly as a variance component, the slope does move between races, by an amount the threshold cannot see.

An audit against practice must hold the team's committed strategy fixed. A planner that stretches its second stint to the flag answers a different question from a team that has already planned a second stop, and the difference looks like a disagreement of +10 laps. With the team's next stop held fixed, the planner and the team agree to within a few laps, out of sample as well as in it. That agreement, however, comes from the horizon: a rule that uses no fitted quantity and simply balances tyre age before the team's next stop does as well.

The stops carry what the fit misses. Treating the stop as an optimal-replacement decision, the lap a team chose implies a ratio of the new tyre's wear rate to the old one's. That implied ratio is a ratio of stint lengths and uses no fitted slope, and it follows compound hardness in the direction physics predicts, in both directions of change, where the fitted slopes are no better than chance and, given a baseline pace per compound, only partly better: teams act on a difference between compounds that the fitted model resolves only in part.

The modelling is simpler than much of the work cited, and it ignores energy management. The contribution is the protocol and what it returned.

2 Related work

Race-strategy research is mostly about building the optimiser: discrete-event simulation, Monte Carlo, dynamic programming, mixed-integer programming, reinforcement learning and game theory, each used to compute a better plan than a team would otherwise choose.

Bekker and Lotz [2009] give the earliest sustained operations-research treatment, simulating a full race with overtaking and traffic under the refuelling rules of the time. Heilmeier et al. [2018] introduce the lap-time decomposition much of the later work rests on, combining fuel mass, tyre degradation and driver effects, and Heilmeier et al. [2020a] extend it to a probabilistic simulation sampling accidents, safety cars and lap-time variability. Carrasco Heine and Thraves [2022] give an exact dynamic program over stop laps and compound choice, and Aguad and

Thraves [2024] formulate the problem as a zero-sum feedback Stackelberg game between two drivers. In endurance racing the same machinery has been built almost entirely for electric cars: van Kampen et al. [2022] plan stints under thermal limits, van Kampen et al. [2024] price a competitor's response by model predictive control, and de Vries et al. [2026] pose the race as a two-agent game. Boettinger and Klotz [2023] simulate GT racing at the Nordschleife and learn a stop policy. Recent Formula 1 work adds learning: Thomas et al. [2025] apply explainable reinforcement learning to compound and stop timing, Fieni et al. [2025] combine a mixed-integer program with an RL agent, Fieni et al. [2026] extend it to self-play, and Todd et al. [2025] forecast tyre energy from team telemetry.

Validation out of sample. Several of these works validate something on data they did not fit, and what they validate differs. Cappello and Hoegh [2026] build a Bayesian state-space model of degradation from the same public timing source used here and evaluate it by rolling-origin cross-validation on 19 race sessions of one season; they also report compound-specific degradation differences as not statistically distinct, which is consistent with how little signal I find at scale. Heilmeier et al. [2020b] train neural networks on the real timing of the six seasons from 2014 to 2019 to reproduce the stop decisions of top-ten finishers, and score them on held-out races. Santillana [2026] calibrates a real-time Monte Carlo engine on 126 races and validates its outcome probabilities on held-out seasons, while its degradation models are fitted per race. Tulabandhula and Rudin [2014] forecast the change in rank position over a pit cycle in professional stock-car racing, validate on the later part of each race, and build a tyre-change decision system on those forecasts. What none of them does is hold out whole seasons of a circuit and ask whether a fitted degradation *slope* predicts them, across championships, which is the first question here.

Comparison with what teams did. Heilmeier et al. [2020b] are the closest: their networks learn to imitate real stops, so their held-out score is a comparison with team decisions, though of an imitator rather than of an optimiser. Santillana [2026] compares the stop counts its engine recommends with the strategies teams executed. Bekker and Lotz [2009] compare with real races and Aguad and Thraves [2024] compare strategic with non-strategic agents inside their own game. What the audit here adds is lap-level: an optimiser's recommended lap for each of a large sample of real first stops, against the lap the team chose, with the team's committed strategy held fixed.

The stop as a replacement decision. A pit stop replaces a wearing asset at a fixed cost, which makes it an optimal-replacement problem in the sense of Rust [1987], who recovered the cost parameters a bus-fleet manager acted on from his observed engine replacements; Romer [2006] asks the same question of professional football coaches and finds them departing from the value-maximising choice. Section 6.4 takes the first step of that programme: it recovers from teams' stops the relative wear rate they act on and compares it with the fitted one.

A mechanism that predicts early stops. Salminen [2026] solves the two-car one-stop game analytically and derives the earliest lap on which an undercut cannot be answered, the only published mechanism I have found that predicts stopping *before* the time optimum. Section 6.4 tests it on the same decisions.

3 Data and protocol

Terms. A *stint* is the run of laps a car does on one set of tyres. For each race the tyre supplier brings three dry *compounds*, from soft (faster when new, wears faster) to hard, and in a dry Formula 1 race every car must use at least two of them, so at least one stop is compulsory. The *pit loss* is the time a stop costs against staying on track. A *neutralisation* (a Safety Car or Virtual Safety Car in Formula 1, a full-course caution in endurance) slows the whole field, which makes a stop taken under it cheaper. An *undercut* is stopping before a rival so that the pace of the new tyres gains the position when the rival stops in turn.

3.1 Sources and scope

Formula 1 lap timing, tyre compound, track status and weather come from the FastF1 API. WEC, IMSA and ELMS timing comes from an endurance database published on Hugging Face as `tobil/imsa` under the MIT licence; it carries lap times, stint structure, flag state and an estimated tyre age, but *not* tyre compound. The endurance degradation models are therefore net-slope models where the Formula 1 one is per compound. Two further Formula 1 sources serve Section 4.3 only: the compounds Pirelli nominated for each race, and second-practice laps from the OpenF1 API. Formula 1 is modelled on the seasons before the 2026 rules throughout; the 12 races of 2026 already ingested are held out of every model, and are audited only in Section 8, separately. Table 1 gives the scope and Table 2 every decision set the results are computed on.

Table 1: Scope. A race-season is one class at one circuit in one season, the unit everything is fitted and reported on.

	Classes	Circuits	Race-seasons	First stops replayed	Seasons
Formula 1	1	25	92	357	2022–2025
IMSA	3	13	136	632	2021–2026
ELMS	2	9	42	171	2021–2025
WEC	1	11	27	120	2022–2026

Which laps are modelled. A lap enters a fit only if it is a green-flag racing lap: in-laps, out-laps, neutralised laps, laps the source marks inaccurate or deleted and wet-compound laps are dropped, so every coefficient is a dry-weather quantity. Laps slower than a fixed multiple (1.10) of the driver–race's own median are trimmed as traffic, and a stint is modelled only if enough laps survive to fit a shape to. The per-circuit counts at every stage are generated into the degradation reports. The frame does not remove variation the filters cannot see: a driver lifting to manage a tyre, an engine mode, a blue flag that cost less than the trim.

3.2 The degradation model

For driver–race i and lap ℓ, with tyre age a and compound c,

$$y_{i\ell} \;=\; \alpha_i \;+\; \phi\,\ell \;+\; \sum_{k=1}^{d} \beta_{c,k}\, a_{i\ell}^{k} \;+\; \varepsilon_{i\ell}, \tag{1}$$

where y is lap time, α_i a driver–race fixed effect absorbing car, driver and the day's conditions, and $\phi\ell$ a lap-number term carrying the fuel burned and any steady track evolution. The degree $d \in \{1, 2\}$ is chosen per circuit by cross-validation. The fuel and wear terms are identified only because different stints start at different laps on fresh tyres; within a stint a and ℓ advance together. In endurance, where cars refuel, the fuel load resets at every stop and the two cannot be separated at all, so the model keeps a single net slope of lap time on tyre age.

Estimation is ordinary least squares. Every published Formula 1 coefficient was checked for estimability: deleting its column lowers the rank of the design for all 116 of them. Standard errors are cluster-robust (CR1) on the driver–race and referred to $t(G-1)$ [Cameron and Miller, 2015]. Drivers in one race also share shocks the driver–race cluster does not cover, and clustering on the race instead leaves 40 of the 73 slope intervals containing zero rather than 16. Section 4.2 estimates that race-level variation directly.

The model gives each compound its own wear slope but no baseline of its own: once the

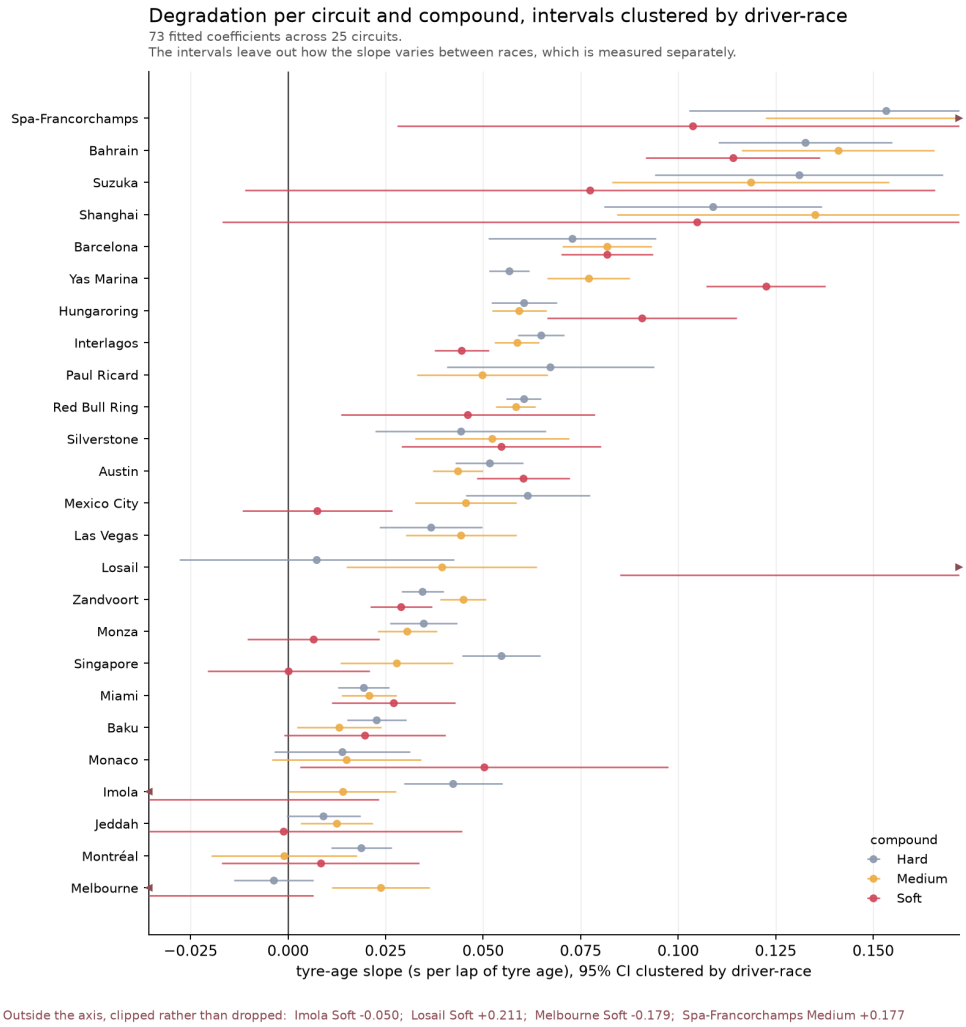


Figure 1: Linear tyre-degradation coefficients for Formula 1, by circuit and compound, with driver–race-clustered intervals: 73 coefficients across 25 circuits. These intervals do not include the variation of the slope between races, which Section 4.2 measures. Points outside the plotted range are named under the axis.

driver-race intercept is removed, compounds are assumed equally fast when new. Section 6.4 tests what a baseline per compound changes.

Two features of Figure 1 matter for what follows. 22% of the slopes have an interval containing zero, and within a circuit 75% of compound pairs have overlapping intervals: the compound distinction the model carries is largely unresolved by the data. Neither is screened out, because screening on resolved slopes would select the circuits likeliest to transfer.

3.3 Transfer, and what limits its measurement

Fit on every season of a circuit-class but one, predict the held-out season, and score the within-stint demeaned residual:

$$R^2_{\text{within}} \;=\; 1 - \frac{\sum_s \sum_{\ell \in s} \big(\tilde{y}_\ell - \tilde{\hat{y}}_\ell\big)^2}{\sum_s \sum_{\ell \in s} \tilde{y}_\ell^{\,2}}, \qquad \tilde{x}_\ell = x_\ell - \overline{x}_s. \tag{2}$$

Demeaning each stint removes the driver–race intercept, which cannot be known for a race that has not happened, so the score tests the *shape* a stint takes. Because a and ℓ advance together inside a stint, that shape is the net of wear and fuel: in Formula 1, $(\beta_c + \phi)$ per lap. It is this net quantity Result 1 is about, and its size limits what any score can show.

The noise ceiling. A low score has two explanations: the slope moved between seasons, or there was never enough signal in the laps for any slope to be recovered. To separate them, lap times are rebuilt from each circuit's own fit with *one* degradation slope for every season, so the slope is stable by construction, plus residuals resampled in whole stints to keep their within-stint correlation. The residuals come from a fit that gives each season its own slope, so real movement of the slope is not counted as noise; with pooled-fit residuals instead the ceiling is lower by at most 0.037 at any circuit. Running the real protocol on the rebuilt laps gives the ceiling: the score a perfectly stable slope earns against the noise present. A circuit is then classified by the ceiling and by the gap between ceiling and measured score: *no signal* if the ceiling is below 0.05; otherwise *stable* if the gap is below 0.03, *unstable* if it exceeds 0.10, and *borderline* between. The ceiling needs at least three seasons, which Paul Ricard and Shanghai lack, and the synthetic fits for Austin and for WEC Hypercar at COTA are degenerate and excluded. These verdicts are descriptive thresholds, not tests.

Between-race variance. The ceiling says whether the score could detect movement; it does not measure the movement. For that, each circuit-compound gets one linear slope per season, each season being a single race, with its driver–race-clustered standard error. The between-race variance τ^2 of those slopes is estimated by restricted maximum likelihood and by the method of moments [DerSimonian and Laird, 1986], with a Q-profile interval [Viechtbauer, 2007], and summarised by Cochran's Q and by I^2, the share of the observed spread not due to sampling error [Higgins and Thompson, 2002]. Because the per-race standard errors do not allow for shocks common to every car at the same moment of a race, Q, τ and I^2 are if anything overstated. The same is done for each endurance circuit-class with its net slope.

3.4 The planners

The multi-stop dynamic program. For an endurance race of L laps with green pace p, net slope β, pit loss π and fuel range F, a plan is a partition of L into stints $(n_1, \ldots, n_{m+1})$ with m stops, of total time

$$T(n_1, \ldots, n_{m+1}) \;=\; m\pi \;+\; \sum_{j} \left(n_j p + \beta \, \frac{n_j (n_j + 1)}{2} \right), \qquad n_j \le F, \tag{3}$$

minimised exactly by a dynamic program in remaining laps for each m from the fuel minimum $\lceil L/F \rceil - 1$ upward. A race is *tyre-limited* when the time-optimal m exceeds that minimum.

The Monte Carlo engine. The single-next-stop engine evaluates every candidate lap for the next stop over 5,000 shared draws. Each draw samples the degradation and fuel coefficients from their sampling distributions ($t(G-1)$ scaled by the cluster-robust standard errors), the neutralisation rate from its Gamma posterior, and lap noise at the cross-validated residual scale. Candidate laps share draws, so a comparison between two laps carries no unrelated sampling noise. A *cover-aware* variant runs the audited car and one rival under the same draws, resolves the pit exchange on the lap both have stopped, lets a gained position survive with the circuit's measured rate of adjacent-car swaps, and lets the rival answer each of the audited car's laps with its best reply: a Stackelberg leader in the sense of Aguad and Thraves [2024], solved by enumeration.

3.5 The audit

Decisions. The engine of Section 3.4 is the planner audited. For every race with a fitted model, the first stop of each of the top 5 classified finishers in Formula 1 (5 cars per class in endurance) is replayed. The engine is asked 5 laps before the real stop in Formula 1 and 3 in endurance, given the state as it was, with rivals on their real plans. Stops before lap 8 (6 in endurance) are excluded as incidents. This is a replay, not a forecast: the decision point is defined by a stop that happened.

The committed horizon. The engine plans one stop. If it is allowed to run the stint after that stop to the flag, it answers a different question from a team that has already planned a second stop, and will stop later for that reason alone. Every decision whose driver stopped again is therefore replayed with the race ending, for the engine, at that driver's real next stop. Everything after that stop runs on fresh tyres whichever first-stop lap is chosen, so it is common to every candidate: the optimum over the shortened horizon is the optimum given the team's own next stop. In endurance each stint is also bounded by the fuel range. Run to the flag instead, the same engine is a median +10 laps late in Formula 1; that number is a property of the question, and Section 6 reports the committed replay.

Out of sample. The circuit models are fitted on the seasons the decisions come from. Every committed decision is therefore also replayed with the fuel and degradation coefficients refitted on every pre-2026 season except its own, at the published degree; a circuit raced in one season only cannot be replayed this way and is counted out.

Reference rules. With one stop before a fixed endpoint and linear slopes, the time-optimal stop k laps after the decision solves

$$\beta_{\text{old}}\,(a_0 + k) \;=\; \beta_{\text{new}}\,(W - k), \tag{4}$$

for a tyre aged a_0 and W laps to the endpoint. It depends on the slopes only through their ratio; the fuel term and any cost paid once per stop cancel. At a ratio of one it balances tyre age either side of the stop and uses no fitted quantity at all; I call that the *ratio-one rule*. Three further rules are scored on the decisions run to the flag: B1 divides the race into equal stints, one stop in Formula 1 and the fuel minimum in endurance; B2 stops when the stint's accumulated loss

$\beta\, a(a+1)/2$ first exceeds the pit loss; B3 runs the fuel load out, and is undefined in Formula 1, which has not refuelled since 2010.

What the stops reveal. Equation 4 can be inverted at the lap the team chose, giving the ratio $\beta_{\text{new}}/\beta_{\text{old}} = (a_0+k)/(W-k)$ that would have made that lap optimal: the age of the tyre at the stop over the length of the stint after it. No fitted slope enters it. It is an implied parameter, which may stand for what teams know or for what they believe, and it uses one lap per decision. A discrete-time hazard uses every lap of the stint instead: the probability of stopping on lap t, given the car is still out, is modelled as a logistic function of the old tyre's next-lap loss $\hat{\beta}_{\text{old}} a_t$ and the new tyre's loss to the endpoint $\hat{\beta}_{\text{new}}(W-t)$, with the fitted slopes plugged in and an indicator for a neutralised lap. The ratio of the two weights is one if teams weigh the tyres as the fit does. Inside one stint $a_t + (W-t)$ is constant, so the two weights are told apart only by how the fitted slopes and W vary between stints; the slopes' own noise is therefore part of what identifies the ratio, and every bootstrap replicate refits them on the races it drew. W is the team's next stop, known only afterwards.

Intervals. Unless stated otherwise, every interval is a percentile bootstrap at the level the data vary at: circuits or circuit-classes for transfer, races for the audit, since drivers in one race share a track, a Safety Car and a tyre allocation.

Table 2: Every decision set a result is computed on. All Formula 1 sets are drawn from the seasons before the 2026 rules.

Decision set	n	Used for
All first stops replayed, four championships	1,280	Figure 4
scored by B1–B3	1,263	Table 3
Formula 1, top five, asked five laps ahead	357	run to the flag
replayable with the next stop fixed	352	committed replay
one-stop races	150	residual
refitted without the season	347	out of sample
between two compounds, linear circuits	232	implied ratio, hazard
Formula 1, every finisher, five laps ahead	1,152	design checks
Endurance, replayable with the next stop fixed	784	committed replay
Formula 1, first stops with no neutralisation in the window	213	cost of the residual

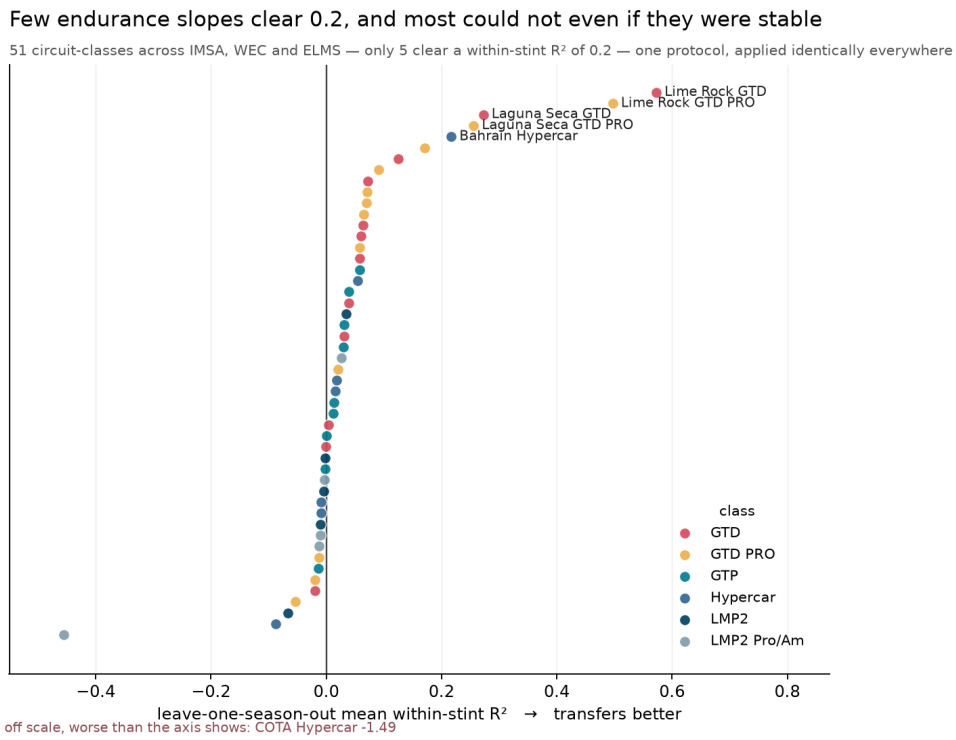


Figure 2: Leave-one-season-out within-stint R^2, one point per endurance circuit-class. Of 51 measured, 5 clear 0.2.

4 Result 1: transfer is mostly not measurable, and slopes do move

Measured as a threshold count, transfer looks rare. Of 51 endurance circuit-classes, 5 clear a within-stint R^2 (Equation 2) of 0.2 (Figure 2); the best is +0.573 at Lime Rock GTD, and almost everywhere else a slope fitted on the other seasons predicts the held-out one no better than that season's own mean. Formula 1 looks the same: 3 of 22 circuits clear it. The rest of this section shows that the count is mostly a statement about signal, not about stability.

4.1 Why the score is low: fuel and wear cancel inside a stint

Over the 22 Formula 1 circuits with a ceiling verdict, the median lap-number term is -0.053 s per lap and the median wear slope +0.040: within a stint the car gains almost as much from burning fuel as it loses to its tyres, and the net slope the score has to predict has a median size of 0.030 s per lap of tyre age (the median of its absolute value per circuit; medians do not add), against a median lap noise of 0.67 s. At Zandvoort the two are -0.042 and +0.034. The size of the net slope ranks the circuits' ceilings with a Spearman correlation of 0.83.

The noise ceiling of Section 3.3 makes this precise (Figure 3). It runs from +0.011 at

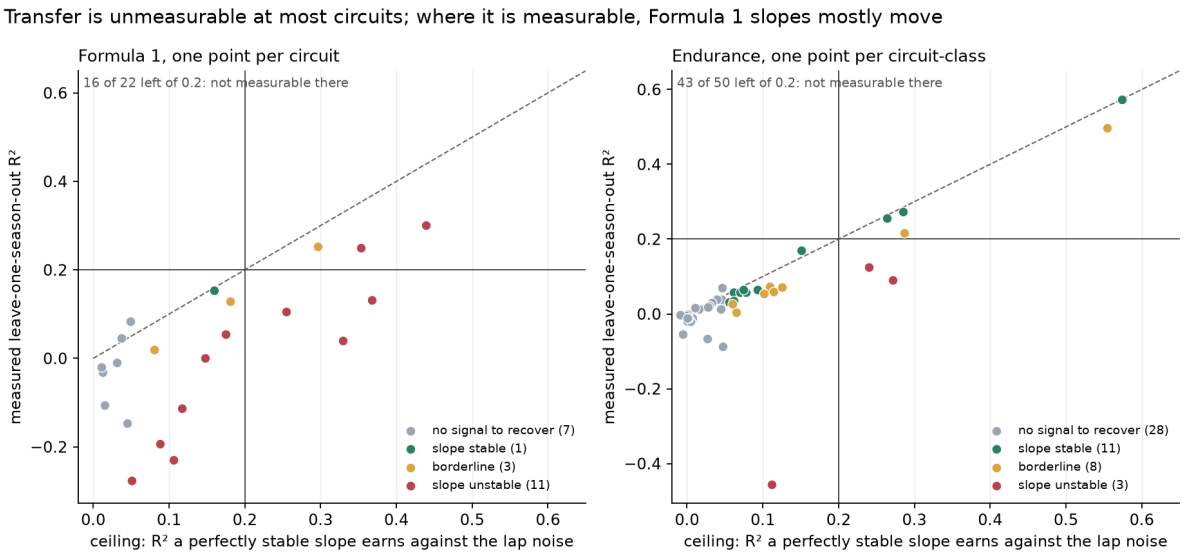


Figure 3: Measured leave-one-season-out score against the ceiling a perfectly stable slope earns against the same lap noise. Points near the dashed diagonal transfer as well as a stable slope could; points well below it have slopes that move. Left of the vertical line, no slope could have cleared 0.2.

Zandvoort to +0.439 at Jeddah, and at 16 of 22 Formula 1 circuits and 43 of 50 endurance circuit-classes a perfectly stable slope could not have cleared 0.2. The threshold count therefore measures how much signal a circuit carries at least as much as it measures stability.

Priced against its ceiling, Formula 1 has 11 circuits whose slopes move, 1 whose slope holds, 7 with no recoverable signal and 3 borderline. The endurance classes divide differently: 28 with no signal, 11 stable, 8 borderline and 3 unstable. Whether that contrast belongs to the championships or to the model, which carries a slope per compound in Formula 1 and one net slope in endurance, these data cannot say.

4.2 How much the slope moves between races

The ceiling tells whether the score could have seen movement; the variance components measure it, on the linear form of Equation 1 fitted to one season at a time. Over 45 Formula 1 circuit-compounds with at least three seasons, Cochran's Q rejects a common slope at 31 of them, and the median I^2 is 0.84 [0.66, 0.91]: most of the spread between seasons is not sampling error. The between-race standard deviation is a median 33% [19, 59] of the circuit-compound's mean slope by restricted maximum likelihood (REML), 31% by the moment estimator, which runs low with three or four seasons, and its interval excludes zero at 30 of 45. Estimated as one relative standard deviation common to the 38 circuit-compounds whose mean is resolved from

zero, in a hierarchical model with one mean each, it is 62% [28, 96]. The two summaries differ by construction: the common value weights every season's slope, so the circuit-compounds that move most count in full, while the median is taken over per-circuit estimates of which several are zero, as REML returns with three or four seasons. I^2 is highest where the ceiling verdict says unstable (median 0.87) and lowest where there is no signal (0.63), so the two measures agree where both can speak. In endurance, Q rejects at 42 of 51 circuit-classes with a median I^2 of 0.87, including 0.88 among those the ceiling calls stable. Slopes move between races at most circuits; at most circuits, too, the within-stint score is too weak an instrument to notice.

Three objections do not change this. *Compound labels*: coded by the real compound each lap ran on, Q rejects at 31 of 43 and the median share is 33% [23, 65]; at the 7 circuits whose nomination never changed, at 10 of 15, with a median of 33% [13, 89]. A change of rubber is not what moves the slope. *Net slope*: season by season, the wear slope and the lap-number term trade off against each other, so the quantity the score actually measures is their sum, estimated directly by writing $\phi\ell + \beta a = \phi(\ell - a) + (\beta + \phi)a$. It moves more: Q rejects at 40 of 45, with a median I^2 of 0.91 and a between-race standard deviation of 97% [70, 163] of its mean. *Estimator*: the shares above are REML; the moment estimator gives lower values (31% for the published coding, 80% for the net slope), as it should with three or four seasons.

Track temperature. The first candidate cause is the track. Regressing the season slopes on the race's median track temperature, with one intercept per circuit-compound, gives +0.010 s per lap per ten degrees (standard error 0.004), against a median within-circuit range of 11 degrees between seasons. The effect is real and small: it removes 2% of the between-race variance.

This variance also bears on the audit. The random-effects standard error of a circuit-compound's mean slope, which counts the between-race variation, is a median 1.8 times the driver–race-clustered one the engine draws from, and up to 8.6 times. Section 6 reruns the audit with the wider distributions.

4.3 Repairs a team would try

Code the tyre by what it was. Compound labels are relative: Pirelli nominates three of its compounds for each race, so the same circuit's Medium can be a different rubber from one season to the next. I collected the nomination for all 92 races from 2022 to 2025 from the race articles of Wikipedia, checked against Pirelli's press releases where they state it in text (50 races,

agreeing every time), and recoded every lap. Where the allocation changed, at 15 circuits with a ceiling verdict, the median score moves from 0.005 to -0.012, a paired difference of -0.012 [-0.349, +0.001], better at 4 circuits and worse at 11. Recoding gives each real compound fewer seasons than its relative label had, and does not repair transfer.

Measure it on Friday. In second practice every car runs long runs on that weekend's tyres and asphalt. Used in place of other seasons, the Friday slope scores -0.502 against 0.021, worse in 44 of 56 races, a paired difference of -0.541 [-1.127, -0.236]. Friday runs degrade faster than the race by a median +0.059 s per lap; removing that offset, learned on the other circuits, still leaves -0.339.

Price the temperature. Adding tyre age times track temperature to each circuit's own model, with the held-out race's temperature, changes the score by a median -0.065 [-0.192, -0.042]: three training races cannot pin down a temperature sensitivity. Borrowing the sensitivity from the other circuits instead changes it by +0.007 [-0.001, +0.019], better at 15 of 22 circuits, too little to establish.

Baselines. Predicting each season only from earlier ones, the question a team actually faces, costs a median -0.046 [-0.114, +0.005] against leave-one-season-out. A slope common to the category, the median of every other circuit's fit, scores -0.034 [-0.099, +0.046] against the circuit's own: over all circuits a circuit's history does not measurably beat a slope that knows nothing about it. At the 15 circuits whose ceiling exceeds 0.05, the circuit's own slope recovers a median 0.42 of its ceiling against 0.23 for the category slope, a paired difference in score of -0.047 [-0.178, +0.058] that is not established either. Refitted with the traffic trim at 1.05 or 1.20 instead of 1.10, at most 18% of slopes move by more than a tenth of their value and the median score ranges from 0.017 to 0.051: the low scores are sensitive to the trim, and no setting brings them near the threshold.

4.4 Car type

Grouped by car type, the raw scores favour GT3 classes, +0.104 against -0.060 for prototypes, a difference of +0.164 [+0.059, +0.312] with a permutation p of 0.0011. That comparison treats classes at different events as exchangeable, and they are not: the two highest GT3 scores, both at Lime Rock, come from an event with no prototype class. Paired within the 9 events that

run both, GT3 scores higher by +0.035 [-0.005, +0.085], at 5 of them, with an exact sign-flip p of 0.22. The raw difference is mostly a difference between circuits, and I do not claim one between car types. For the same reason the near-specification ELMS LMP2 field cannot speak to Balance of Performance, the adjustments that equalise different cars in one class: at 4 of its 5 circuits its laps carry no recoverable signal.

5 A note on the endurance planner

The dynamic program of Equation 3 also reports, for each endurance race, whether the time-optimal plan needs more stops than the fuel minimum: 25 of 205 race-seasons do. The regime is decided inside the planner by weighing an extra stop against what the tyre gives back, so it is close to an identity of the model and is reported here only as a description of where these championships sit. Ranked by pit loss alone the race-seasons separate at an AUC of 0.977 [0.955, 0.994], by the net slope alone at 0.865 [0.806, 0.915]; dropping either from the planner's own index, $\log\beta + 2\log F - \log\pi$, costs a similar amount (+0.030 [-0.017, +0.084]), so both set the regime. Refuelling is a median 5% of the stop loss and does not change the count, and no tyre-limited race-season has a pit loss above 22.5 s, an edge set by one race. Figure 5 in the Appendix shows the class-level relation ($r = -0.982$ on six class points). The decision endurance teams more often face, whether to change tyres at a fuel stop, is not in the model (Section 7).

6 Result 2: with the strategy held fixed, the planner agrees with teams, and the agreement comes from the horizon

The planner chooses the lap of the next stop to minimise one car's expected race time, on the decisions and horizons of Section 3.5. It represents neither the tyre allocation, nor traffic, nor energy management, nor a championship position. Every number below measures a disagreement between a model and a practice, not which was faster.

6.1 The agreement

With each team's next stop held fixed, the planner lands a median of +2 laps [+1, +4] from the real stop over the 352 replayable Formula 1 decisions, against +9.5 when the same decisions are run to the flag (Figure 4). In endurance the interval contains zero in all 6 classes.

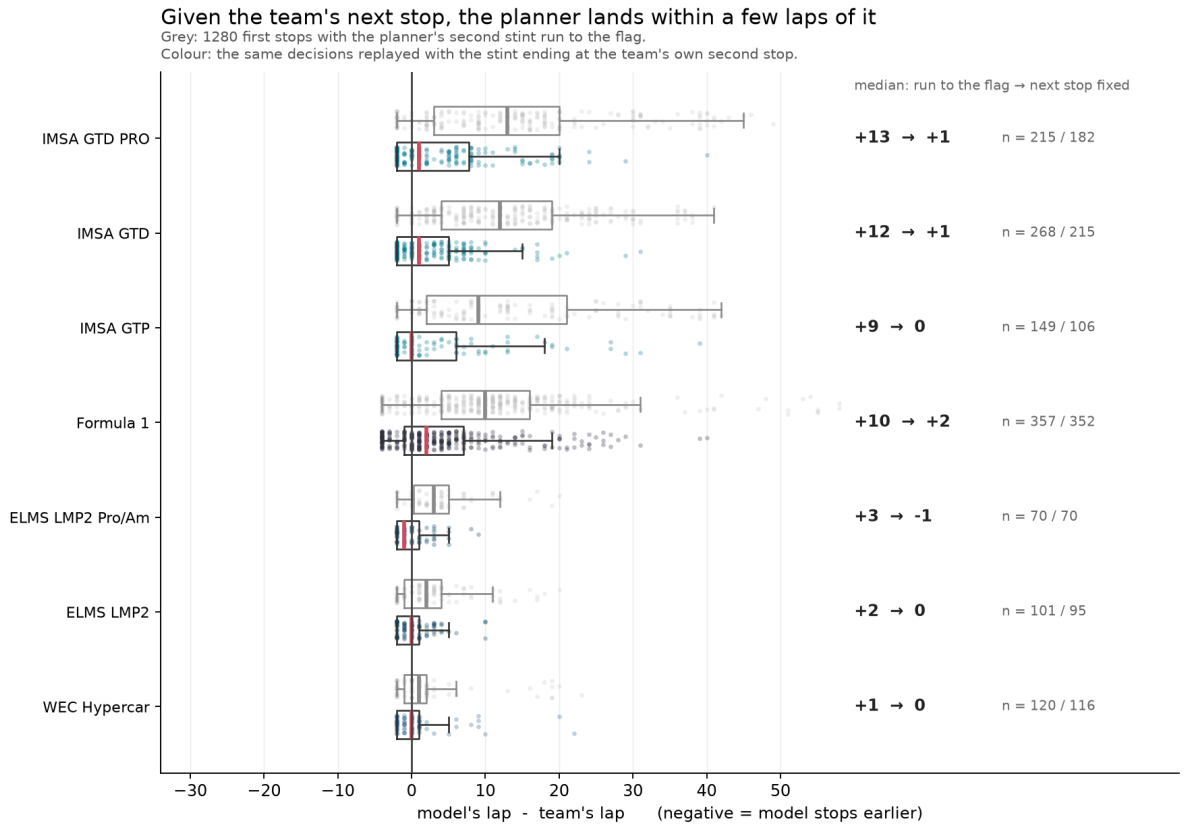


Figure 4: Model lap minus team lap over the 1,280 replayed first stops. Grey: the engine run to the flag. Colour: the same decisions with the stint ending at the team's own next stop, for the 1,136 decisions that can be held to one (Table 2). Boxes show quartiles.

The agreement survives the checks a reader would ask for first.

- *Out of sample.* With slopes fitted without the decision's season, the median is +3 [+2, +4] over 347 decisions, against +2 in sample on the same ones; the recommended lap moves for 64% of decisions, by a median of 1 lap.

- *Wider uncertainty.* With every slope's sampling distribution widened to its random-effects standard error, the median is +3 [+1, +4].

- *Design.* Asked ten or fifteen laps ahead instead of five, and on every finisher instead of the top five, the median runs from +1 to +3 across six designs.

- *Green flag only.* On the 291 decisions whose real stop was under green, it is +2 [+1, +3]; keeping only decisions whose *next* stop was under green, +2 [+1, +2]. In IMSA, where 53 to 56% of first stops are taken under caution, the green-flag medians are 0, 0 and +0.5 for GTP, GTD and GTD PRO.

- *Not a narrow window.* In Formula 1 races with a second stop, the window between the decision and that stop is a median 24 laps; a lap drawn uniformly inside it would miss the real stop by a median 8.3 and the window's midpoint by 7.5, against 3 for the engine on the same decisions. In endurance the engine beats both only in the 3 IMSA classes; in WEC Hypercar and ELMS the fuel range leaves 4 to 7 laps and the replay is uninformative.

Conditioning on the team's next stop is conditioning on a later decision, made knowing where the first stop had been. The committed median is agreement given the plan the team revealed, which is what makes the planner and the team answer the same question; it is not a measure of how good the planner is.

6.2 Where the agreement comes from

Equation 4 describes what the engine is doing: it lands a median of 0.9 laps from the engine's own lap on 243 decisions at linear circuits. Its ratio-one form, which uses no fitted quantity, is the natural reference, and it ties the engine's median absolute error in 7 of 7 classes (Table 3). The agreement of the committed replay is therefore produced by the horizon: once the team's next stop is known, balancing tyre age before it is as close to practice as the full engine. On this measure the Monte Carlo, the neutralisation hazard and the fitted slopes add nothing measurable to the agreement.

In endurance this is forced. The model has one net slope, so $\beta_{\text{old}} = \beta_{\text{new}}$, the timing reduces to equal tyre ages either side of the stop whatever the slope is, and the replay tests the fuel and neutralisation logic rather than the tyre model.

Table 3: Median absolute lap error against the real stop, per class. The first four columns are run to the flag, on the 1,263 decisions the rules can score; the last two are given the team's next stop, on the decisions the committed replay can make. Bold marks the best of the first four.

	Run to the flag				Next stop fixed	
	Optimiser	B1	B2	B3	Optimiser	Ratio one
Formula 1	10	11	**7**	–	4	4
IMSA GTP	9	9	54	14	2	2
IMSA GTD	12	**7**	18	14	2	2
IMSA GTD PRO	12	**9**	33	15	2	2
WEC Hypercar	2	2	39	**1**	1	1
ELMS LMP2	**2**	3	45	4	2	2
ELMS LMP2 Pro/Am	3	**2**	45	4	2	2

Run to the flag, a rule of thumb sits closer to practice than the optimiser in 5 of 7 classes, and in 3 of them the rule is B1, which uses no fitted quantity. Given the team's next stop, the optimiser is closer than every rule in 5 classes and ties the best in 2, but the ratio-one rule, given the same next stop, does as well. The rules' advantage over the optimiser run to the flag came from the question that optimiser was asked.

6.3 The residual

What remains is concentrated in one-stop races, where the planner and the team were always answering the same question: +4 laps [+2, +7] over 150 decisions, +5 [+2, +9] out of sample. Part of it belongs to the audit's window. Asked five laps before the stop, the planner can be at most four laps early and arbitrarily late, and 13% of committed decisions sit on that bound; replayed for every finisher fifteen laps ahead, the one-stop median is +1 [-1, +3]. Part belongs to extrapolation: 31% of one-stop recommendations stop at a tyre age beyond the 95th percentile of the stints observed on that compound at that circuit, where the linear slope is no longer measured, and where the recommendation stays inside that range the one-stop median is +3. What it costs is taken up in Section 6.5.

6.4 What the stops reveal

The residual has a form that does not depend on the window. Inverting Equation 4 at the lap each team chose, over the 232 decisions between two different compounds at linear circuits, gives a median ratio $\beta_{\text{new}}/\beta_{\text{old}}$ of 0.77 [0.70, 0.83] against 1.02 fitted. Because the implied ratio is $(a_0+k)/(W-k)$, this says first that teams' first stints are shorter than the stints that follow them; the fitted value it is compared with is a ratio the fits do not resolve, since 75% of compound pairs overlap.

The direction of the change separates the two readings. If teams act on a difference between compounds that the fits miss, the implied ratio should fall below one when they move to a harder compound and rise above one when they move to a softer one; if something else shortens first stints, it should fall below one either way. It splits by hardness (Table 4): to a harder compound the implied ratio is 0.73 [0.68, 0.77] over 204 decisions, to a softer one 1.74 [1.43, 2.53] over 28 decisions in 19 races, and it lies on the side of one that hardness predicts for 79% of decisions. The fitted ratios do so for 48%, no better than a coin: they put the harder compound at 1.02 of the softer one's wear and the softer at 0.87 of the harder one's. The implied ratio therefore measures what the fit does not resolve: teams stop as if they know which compound lasts, and the published fit does not show it. Part of this is the same knowledge exercised the other way round, choosing the harder compound for the longer stint, which the inversion cannot tell from a wear rate. The pooled 0.77 is mostly the harder-compound case, which is the common one.

Part of what the fit misses is its own specification. Refitted with one baseline pace per compound, the harder compound is slower when new, by +0.115 s [+0.045, +0.206] over the decisions that move to it, and the refitted slopes order the compounds on the expected side for 64% of decisions instead of 48%: better than chance, still short of the teams. With those baselines in the time-optimal single stop, the planner's lap moves from +3 to +2 laps [+1, +4] of the team's, so the shorter first stints remain.

Table 4: Implied and fitted ratios $\beta_{\text{new}}/\beta_{\text{old}}$ by the direction of the compound change, Formula 1 decisions between two different compounds at linear circuits. Intervals resample races.

Change	Decisions	Races	Implied ratio	Fitted ratio
To a harder compound	204	50	0.73 [0.68, 0.77]	1.02 [0.93, 1.14]
To a softer compound	28	19	1.74 [1.43, 2.53]	0.87 [0.57, 1.36]

The stop hazard, fitted on all 3,500 laps those 232 first stints were at risk, puts the weight teams give the new tyre's fitted loss, relative to the old tyre's, at 0.50 [0.34, 0.77] with the

slopes refitted in every replicate ([0.32, 0.70] if they are held fixed). It agrees with the implied ratio in sign, not in size, and it cannot separate the two directions: 0.65 [0.47, 0.87] to a harder compound, 0.56 [0.29, 1.45] to a softer one. Since it is identified only across stints, through slopes that do not resolve the compounds and an endpoint known only afterwards, I read it as consistent with the implied ratio rather than as a second measurement of it.

Compound knowledge is not the only thing the implied parameter can absorb: wear that accelerates late in a stint, the value of keeping the option of a Safety Car and tyre management to a planned stint length all act on the same ratio. Five further mechanisms I could test do not account for the shorter first stints:

- *Wear that grows with fuel load.* A term in tyre age times lap number has mixed sign, positive at 10 of 19 circuits, and rerunning Equation 4 with it moves the median gap from +3 to +4 laps, away from the real stop.

- *Heavier first stints.* Fitted separately, first-stint slopes are steeper than later ones at only 21 of 52 circuit-compounds, and of the 17 differences that are significant, 4 go that way; using them moves the gap from +3 to +4 laps, again away.

- *Track position.* The cover-aware engine moves the recommendation closer to the real stop on 34 of the 150 one-stop decisions and further on 74, and under each of the 4 rival rules below it moves closer on at most 37 (sign test $p \leq 0.006$).

- *The earliest undefendable undercut* of Salminen [2026], with the position-keeping defence, predicts no undefendable undercut before the flag on 72 of the 109 one-stop decisions carrying a prediction. On the 37 where its lap falls inside the race, its absolute error exceeds the single-car optimiser's by a median +9 laps [+1, +15], closer on 9 and further on 25. Across all Formula 1 decisions, for it to reproduce what teams do, tyres would have to wear at a median 0.185 s per lap, a median of 3.4 times the rate measured for the same decision.

- *A gain at the stop.* With the number of stops fixed, any time gained or lost once per stop is common to every candidate lap and cancels; it cannot move the lap. The data show such a term with the opposite sign anyway: laps on a new set come in slower than the fit expects, which is tyre warm-up.

If the fitted slopes understated wear, the planner would stop late, as it does. Re-estimated

from an independent lap history in the Ergast schema, distributed on Kaggle, which separates wear from fuel by a different method, they differ by a median +0.0006 s per lap over 23 circuits, against a median wear slope of 0.046, so no such bias shows. A full structural estimate in the manner of Rust [1987], with the compound difference as a parameter rather than a fitted input, is where explaining the ratio would start.

6.5 What the residual costs

The planner prices the residual at tenths of a second. Over the same stretch of each real race, from the decision to the next stop or the flag, with neutralised windows dropped, I compared each car's elapsed time with what its own median green lap predicts, within races and controlling for the window's length. Over 213 first stops in 44 races, each lap of distance from the planner's lap goes with 0.42 s of extra time [-0.24, +1.11], against 0.25 s by the planner's own pricing. Between cars of one race the excess time has a median standard deviation of 5.6 s, so the smallest per-lap cost these decisions could detect with 80% power is about 0.96 s, and detecting the planner's own price would take about 3,100 decisions. This is an association between cars, open to every reason a team had for its lap.

6.6 Which car is the rival

A competition-aware model needs a rival, and the timing does not name one. The published replay pairs each car with its nearest neighbour on classified position, which, as Charles Thraves pointed out, picks the car ahead on some decisions and the car behind on others. Against it I ran his proposal (the car whose standalone optimal race time is closest), the car ahead on the clock that the model of Salminen [2026] is defined on, and every car within one pit loss. Over the 271 decisions where all 4 apply, they name the same car 17 times; pairwise, two rules agree on between 20% and 65% of decisions, a median of 36%. The recommendation barely moves between them: the cover-aware engine stays later than the real stop under every rule. The game-theoretic literature is built on a two-car abstraction [Aguad and Thraves, 2024, de Vries et al., 2026, Salminen, 2026]; measured against real decisions, which two cars those are is under-determined.

7 Threats to validity

The committed horizon is hindsight. The replay of Section 6.1 is given the team's next stop. Restricting to green-flag next stops removes the opportunistic ones, not a plan revised during the race.

The endurance tyre clock. Endurance teams often double-stint tyres, and the decision they face is often whether to change tyres at a fuel stop. The endurance model counts tyre age from the source's estimate, carries one net slope with no compound, and does not represent that decision; its tyre-limited regime answers a question endurance teams rarely ask, and its replay cannot test the tyre model (Section 6.2).

Slopes depend on the strategy. A driver told to make a set last drives it more gently, so a fitted slope depends partly on the stint the team chose, which pushes in the direction of the one-stop residual and would show up between seasons as instability. Public timing cannot remove it.

Reuse, not refitting. As Giona Fieni put it, a model-based policy retunes its degradation curves for each car and circuit and never depends on a slope transferring. Result 1 bounds approaches that carry fitted parameters forward (Section 4); for a design that refits as a matter of course it is an argument for refitting.

Decided after seeing the data. The committed replay, the rival correction, the design checks and the analyses of Sections 4.2, 6.3 and 6.4, including the split of the implied ratio by direction, were all built after the run-to-flag gap was seen. The protection against a forking path is a test the analysis could not have seen, frozen in Section 8.

Withdrawn estimators and thin constants. An estimator of track evolution built for this work was withdrawn after it failed on real races. The best endurance transfer scores rest on two or three folds each. Both are documented in the repository.

8 Reproducibility and a prospective test

Every artifact derived from committed data is regenerated by continuous integration, which fails on any difference between the working tree and the repository. Every number in this

manuscript is a macro generated from the committed artifacts; the source contains no digits of its own. The state the results were computed from is archived on Zenodo, at `https://doi.org/10.5281/zenodo.22726130`. The repository does not redistribute the two third-party exports, both downloaded from Kaggle, that 3 of its layers read: a Formula 1 history in the Ergast schema (lap times, pit stops, results) and a WEC results table. Those layers cannot be regenerated from a clone; their outputs are committed, and the only published result resting on them is the cross-source slope check of Section 6.4.

A prospective test, frozen before its data exist. The committed replay — planner, seeds, lookback, selection of cars and horizon — is frozen in `paper/prospective_protocol.md`, committed before the Formula 1 races it will be scored on had been run, with its hypotheses and prediction intervals stated in advance: a committed median inside [+1, +4.5] laps over the next full season, and, as a secondary test on the races left in the current season, inside [-2, +12]. Two hypotheses added before the first of those races do not depend on the lookback: the whole field replayed fifteen laps ahead, and the implied ratio. The frozen models were fitted on cars built to the previous rules, so the test also asks whether the replay survives a rule change. The result will be reported against the dated file whichever way it falls. One frozen hypothesis fixes the pooled implied ratio, whose value depends on how many changes go to a harder compound; the split of Table 4 came after the freeze and is not part of it.

The 2026 races already ingested. The 12 Grands Prix of 2026 already in the repository were ingested before the test was frozen and are held out of every model in this paper. The audit had never been run on them; their laps did enter an early version of the noise ceiling, before it was restricted to 2022–2025. Replayed with the frozen models, out of sample but *not* pre-registered, the 11 with a replayable decision give a committed median of +1 laps [-4, +4] over 54 decisions, -1 [-6, +3] for the whole field fifteen laps ahead over 121 decisions, and an implied ratio of 0.88 [0.77, 1.09] over 29 decisions, all of them to a harder compound, against 0.73 for that direction in 2022–2025. The committed agreement is not contradicted by the first races under the new rules. The implied ratio sits closer to one on the new cars and tyres, but over 29 decisions its interval reaches the 2022–2025 value, so no change is established.

9 Conclusion

Public timing rarely carries enough signal to say whether a degradation slope transfers, and the reason is physical: inside a stint the fuel burned nearly cancels the tyre's wear, so the shape a stint shows is small against the lap noise. Measured directly, the slope moves between races at most circuits, by a median 33% of its mean, and by about its own size for the net slope a stint shows; neither a change of compound nor track temperature explains much of it. Held to the team's committed strategy, and out of sample, a time-optimal planner stops within a few laps of the pit wall, but so does a rule that only balances tyre age before the team's next stop: the agreement is a property of the horizon. What the stops do reveal is the compound difference the fit misses. Read as an implied wear ratio, the length of a team's first stint against the next follows compound hardness in both directions of change, where the fitted slopes are no better than chance, and only partly better once each compound has its own baseline pace. Teams act as if they know which tyre lasts; public timing, as modelled here, shows it only in part.

Data and code availability

Every quantity in this paper is derived from committed artifacts by scripts in a public repository (Section 8), which also contains the data, the figures and the tests: `https://github.com/mohammedmedjadj/Motorsport-Strategy-Lab`. The state the results were computed from is archived at `https://doi.org/10.5281/zenodo.22726130`.

Acknowledgements

Several researchers replied to cold emails about this work, and their replies changed it. Charles Thraves (Universidad de Chile) identified the rival-selection defect and proposed the standalone-time rule of Section 6.6, and named the endogeneity that bounds it. Wytze de Vries (TU Eindhoven) raised two measurement objections to the endurance regime of Section 5, that class medians pool stops with and without a driver change and that part of the correlation may run backwards; both were tested, and neither disturbs the class ordering. Jorn van Kampen (TU Eindhoven) described the same structure from electric endurance racing, where a tyre change is negligible beside the charging stop. Giona Fieni (ETH Zürich) drew the line between reused parameters and model-based policy stated in Section 7. Felipe Carrasco Heine pointed out that

the lap-time model has no baseline pace per compound, which Section 6.4 tests. None of them is responsible for what this manuscript says.

A The stop cost in the endurance planner

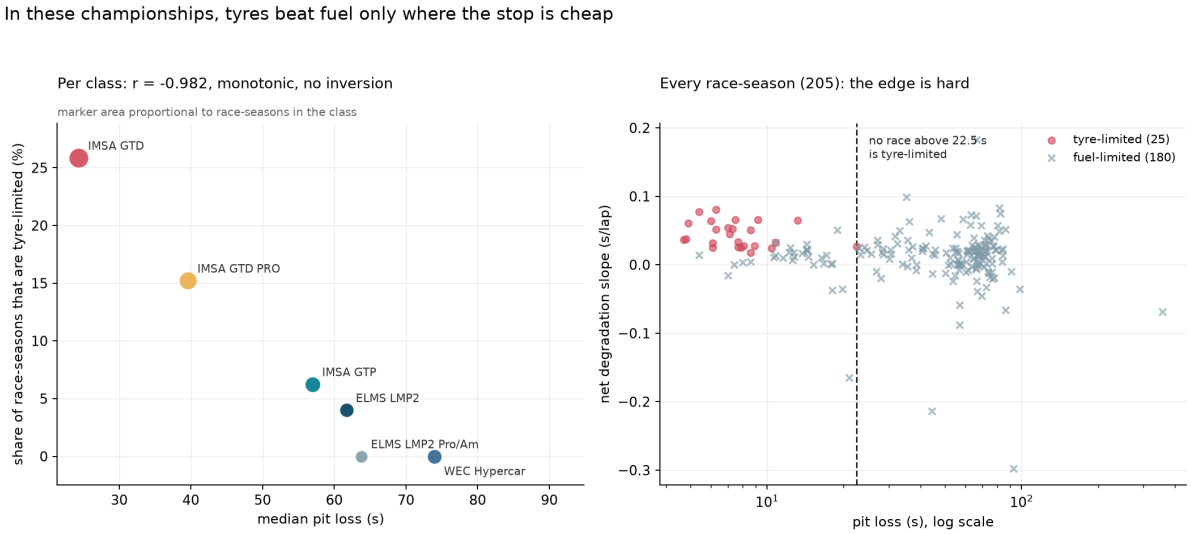


Figure 5: Left: class-level median pit loss against the share of race-seasons whose time-optimal plan needs more stops than the fuel minimum. Right: every race-season, with the empirical edge marked (Section 5).

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